BIO204

Answer 'concisely' for total of 25 marks.

1. Scan the following PSSM matrix in the sequence below and report the top match:
   PSSM matrix: (5)

	1	2	3
A	0.1	0.1	0.7
T	0.6	0.2	0.1
G	0.2	0.5	0.1
C	0.1	0.2	0.1

   Sequence to be scanned: CGATGAGCATTAAAATTCAG

2. What distributions do you expect for thefollowing phenomenon in biology? Highlight the possible rationale and/or significance in each scenario. (10)
   a. Gene density in human genome
   b. Number of genes in mammals
   c. Cell-sizes in a tissue
   d. Number of cells in individual organisms of a species
   e. Food chain network
   f. Number of mitochondria in various cells of an organism.
   g. Bodymass distribution of all animals
   h. Vascular network
   i. Mobility of animals
   j. Number of vertebrae in all mammals

3. A bacterium has higher GC content in the coding regions when compared to noncoding regions. What might explain this? (5)

4. I have a 100 residue long protein sequence. I need to search repeated (interspersed or tandem) triplets of amino-acids. How many combinations I need to search and why? (5)

5. I need to amplify a human gene through PCR. I design PCR primers. How can I ensure that the primers amplify my gene of interest specifically? (5)

6. What are the consequences of most real world networks being scale-free ? (5)

7. Why the thermostability hypothesis for higher GC content of genes is disputed? (5)

8. Why mouse knockouts for many genes do not show any apparent morphological, physiological or behavioural phenotype? (5)

9. Probabilistically infer whether a species is reproductively active or passive by monitoring its daily behaviour across several months. (5)

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25 marks | 1 hour

Answer any five questions in total (drawn from any of the three sections below, but clubbing all questions from the same section into the same section of your answer paper). Please note that each question carries 5 marks.

SECTION A

1) How do we know that auto-regulation is a motif in gene regulatory networks?
2) Why are negative autoregulatory loops selected in evolution?
3) Derive Hill's equation for transcriptional repression.

SECTION B

4) What are the key components involved in the CRISPR-Cas9 genome editing process, and what roles do they play? (4 marks)
5) Describe at least two potential applications of CRISPR-Cas9 each in medicine and agriculture. How could these applications benefit society? (4 marks)
6) What are some ethical concerns surrounding the use of CRISPR-Cas9 technology in human germline editing? (4 marks)
7) Describe the key components of a Yeast Artificial Chromosome (YAC) and explain how it is used for cloning large DNA fragments in yeast cells.? (4 marks)

SECTION C

8) Describe what happens during the first few (4-to-5) steps of a PCR reaction in which one is attempting to amplify a short stretch of DNA from a much longer stretch of DNA, mentioning the number of the cycle in which the very first copies of the desired (shorter) stretch begin to appear in the PCR reaction.
9) Explain why PCR does not progress geometrically throughout all of the tens of cycles of any PCR reaction, mentioning (A) why the underlying reason is not the exhaustion of (i) oligonucleotide primers, or (ii) nucleotide triphosphates, but (B) something else altogether, describing what causes non-geometric amplification.
10) Describe (with accompanying figures) how a site-directed mutation can be incorporated into a stretch of double stranded DNA by using the splicing-by-overlap extension (SOE) PCR technique.
11) Describe at least five of the main trouble-shooting approaches to use when a PCR reaction fails to work, using one line to describe each of the five.